

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter End Semester Examination 2019

Course: Civil Engineering

Sem: III

Subject Name: Hydraulics I

Subject Code: BTCVC303

Max Marks: 60

Date: 14 /12/2019

Time: 10 AM to 1 PM

Duration: 3 Hr.

Instructions to the Students:

1. Solve ANY FIVE questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Solve All Questions in Sequential order.

	(CO)	Marks
Q. 1 Solve the following.	CO 2	12
A) Explain in detail Viscosity, its types and effect of temperature on Viscosity and solve below numerical: Two large Horizontal Plates of infinite extent are 2.5 cm apart and have liquid viscosity of $0.8 \text{ Ns} / \text{m}^2$ filled between them. What force is required to pull a thin plate of negligible thickness and area 0.5 m^2 between two large plates at a speed of 0.5 m/sec . if		06
a) The thin Plate is in the middle of the two large plates?		
B) Derive an Expression for Capillary rise in Glass Tube and solve below numerical: For having blood sample a fine glass capillary of dia. 2 mm was held touching a freshly punctured finger tip. Estimate in 'ml' the volume of blood sample so drawn. Take the density of blood as $1060 \text{ kg} / \text{m}^3$, surface tension as $5 \times 10^{-2} \text{ N/m}$ and its contact angle with glass as zero degrees.		06
Q.2 Solve the following.	CO 2	12
A) Explain and prove Pascal's Law. Also solve below numerical : A Triangular lamina of base and altitude equal to 3.2m is submerged in water vertically. Determine the Total pressure and centre of pressure, when Base of lamina coincides with free surface.		06
B) Derive an expression to find Metacentric Height of floating body by analytical method. Also solve below numerical: A wooden cylinder of diameter 'd' and length '2d' floats in water with its axis vertical. Is the equilibrium stable? Locate the meta-centre with reference to water surface. Specific gravity of wood is 0.60.		06

- Q. 3 Solve the following.** **CO 4 12**
- A) Derive Continuity Equation for three dimensional flows in Cartesian coordinate system. **06**
- B) Water flows through a horizontal pipe of diameter 20 cm with a velocity of 20 m/sec. The Pressure at A and B is given as 25 N/cm² and 20 N/cm² respectively. Find a head loss between A and B. **06**
- Q.4 Solve the following.** **CO 3 12**
- A) An Oil of Specific gravity 0.9 and Viscosity 0.12 Ns / m² flow through a pipe of 2.5 cm diameter with a pressure drop of 0.35 N / cm² in a length of 30m. Determine the discharge ,shear stress intensity at the pipe wall and the power required to maintain the flow. **06**
- B) Crude oil of specific gravity 0.9 and $\nu = 0.15 \text{ cm}^2/\text{sec}$ is flowing through a 25 cm diameter smooth pipe at the rate of 105 LPS. Calculate velocity at a distance of 6 cm from pipe axis, maximum velocity and velocity at the end of laminar sub layer. **06**
- Q. 5 Solve the following.** **CO 3 12**
- A) Define and explain below Dimensionless Numbers- a) Reynolds Number, b) Froude Number, c) Euler Number, d) Mach Number e) Weber Number **06**
- B) The Capillary rise ' h ' of a fluid of density ρ and surface tension σ in a tube of diameter D depends on contact angle θ and gravity g . Prove by Raleigh's method. **06**
- $$\frac{h}{D} = f_n \left[\frac{s}{\rho g D^2} \theta \right].$$
- Q. 6 Solve the following.** **CO 3 12**
- A) Two reservoirs 1 and 2 having level difference between free surfaces of water equal to 5m are connected by pipes AB and BC of diameters 20cm and 30cm respectively. Length of AB is 300m and that of BC is 200m.If flow through the pipeline is 50LPS, Find Loss due to expansion and friction factor f . Draw EG line. **06**
- B) A Siphon Pipe of 20cm diameter 500m long connects two tanks having difference of level in free surface of water of 20m.The summit is at 100 m length from upper tank and 3m above water level of upper tank. Find Q through siphon and pressure at summit .Take $f = 0.02$ **06**

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